MCP WITH AUTOGEN: INTEGRATING AI AGENTS WITH NOTION

Project Overview

In this project, you will tackle the biggest challenge in the current AI landscape: **Interoperability**. Instead of writing brittle, custom API integrations, you will learn to implement the **Model Context Protocol (MCP)**—the new industry standard for connecting AI models to data systems.

You will act as an **AI Integration Engineer**. You will build a custom **MCP Server** using **Flask** that standardizes Notion operations (creating pages, searching databases) into a format that AI can understand. You will then expose this server securely to the world using **ngrok**.

Finally, you will connect this server to a **Microsoft AutoGen** agent team. You will watch as your agents autonomously plan tasks and execute them inside your Notion workspace using natural language, demonstrating a truly decoupled, scalable AI architecture.

What you will Learn?

- **Model Context Protocol (MCP):** Be an early adopter of the standard that is set to replace custom tool integrations.
- **Microsoft AutoGen:** Deepen your knowledge of multi-agent orchestration and tool use.
- **API Development:** Gain proficiency in building lightweight APIs with **Flask**.

- **Tunneling & Networking:** Understand how to use **ngrok** to bridge local development environments with cloud services.
- **Productivity Automation:** Learn the intricacies of the **Notion API**, a valuable skill for building internal enterprise tools.

Why this Project?

- **Future-Proof Skill (MCP):** MCP is being adopted by major players like Anthropic and Replit. Learning this now puts you on the absolute bleeding edge of AI Engineering.
- **Decoupled Architecture:** You learn to build agents that are independent of the tools they use—a key principle of scalable software design.
- **Immediate Utility:** You walk away with a functioning tool that lets you control your personal Notion workspace via chat, which is genuinely useful for productivity.

Module
Understanding MCP
MCP - Application & Idea
MCP as a Solution
MCP Architecture Overview
Before and After MCP
MCP Practical
MCP with AutoGen
MCP in Autogen
MCP in Autogen with Workbench
MCP Practical with Time MCP Server
MCP Workbench in Autogen
Notion Integration
Using MCP with Notion MCP Server
Setting up Notion MCP From Scratch
Hitting Notion MCP Server via Autogen Agent
Flask API & Deployment
Project Architecture and Routing via ngrok
Simple Flask App for Endpoint
Adding ngrok Tunelling to Flask App

Tech Stack & Tools

1. **Protocol:** Model Context Protocol (MCP)
2. **Framework:** Microsoft AutoGen
3. **Backend:** Flask, Python
4. **Networking:** ngrok
5. **Target App:** Notion API